



The Digital Mirage and the Sovereign Oasis: A Theoretical Framework for Infrastructure Sovereignty in Geopolitically Isolated Exclaves

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Abstract: Many smart city initiatives present technological sophistication as synonymous with urban resilience. This paper challenges that assumption by introducing the concept of the “Digital Mirage”, a condition in which the appearance of smart infrastructure conceals structural dependency on external vendors and foreign-hosted data architectures. The paper proposes a transferable theoretical framework, the Sovereign Oasis model, for Infrastructure Sovereignty in geopolitically isolated exclaves, using the Nakhchivan Autonomous Republic, Azerbaijan, as the primary illustrative application context. A Most Dissimilar Systems Design (MDSD) comparative analysis across three cases, Aghali Smart Village as a greenfield reference case, Tallinn, Estonia, as a digital sovereignty benchmark, and Nakhchivan City as a legacy retrofit context, is combined with a process-tracing illustration of the smart grid dependency chain. Drawing on theories of high-modernist planning, polycentric governance, and the political nature of technological artefacts, the paper proposes a three-pillar resilience architecture: a Sovereign Tech Stack maintained through distributed community governance; an Edge Computing model ensuring Graceful Degradation upon link severance; and a Participatory Infrastructure Auditing mechanism grounded in local knowledge. Mean Time to Recovery (MTTR) is proposed as the primary metric for future empirical testing. The Sovereign Oasis model is offered as a design template for exclaves and comparably isolated urban contexts where digital infrastructure dependency constitutes a strategic vulnerability.

Keywords: Smart city; Infrastructure Sovereignty; Exclave urbanism; Digital commons governance; Polycentric governance; Graceful degradation; Open-source urbanism; Nakhchivan

1 Introduction

There is a paradox at the heart of contemporary smart city discourse that becomes visible when examined from the geographic and geopolitical margins of the urban world. Much of the smart city literature focuses on cities embedded in stable national infrastructure grids, with reliable access to international supply chains, legal recourse over vendor contracts, and the institutional capacity to absorb a failed technological experiment without existential consequence.

Geopolitically isolated exclaves, territories administratively part of one state but physically separated from it, represent an undertheorised category in urban studies. Their digital infrastructure challenges are structurally distinct from those of cities embedded in continuous national networks: supply chains are longer and more fragile, external connectivity depends on politically contested transit routes, and the consequences of system failure extend beyond inconvenience into questions of strategic vulnerability. The Nakhchivan Autonomous Republic (Azerbaijan) exemplifies this condition and serves throughout this paper as the primary illustrative context for the theoretical framework developed here.

The dominant model of smart city development, which this paper terms the “Digital Mirage”, is poorly suited to the exclave context. Characterised by centralised data architectures, proprietary vendor ecosystems, and the assumption of stable external connectivity, the Digital Mirage produces urban systems that appear technologically sophisticated while concealing fundamental structural fragility, a concern consistent with critical smart city scholarship on corporate storytelling, datafication, and vendor dependency [1–4]. A city whose water distribution or energy routing depends on a software update from a foreign vendor, or on data processed in an externally hosted cloud server,

is not a resilient city. It is a dependent one. Grossi and Welinder [5] observe that smart city governance frequently prioritises technological maturity over institutional resilience, leaving cities exposed when the conditions enabling that technology are disrupted.

This paper develops a theoretical framework, the Sovereign Oasis model, for Infrastructure Sovereignty in geopolitically isolated exclaves. The framework is organised around a single architectural principle: urban digital systems in exclaves must be designed to function when external links are severed, not merely when they are available. Figure 1 illustrates the conceptual progression from the Digital Mirage condition through the Sovereignty Gap to the three-pillar resilience architecture. The theoretical question driving the analysis is: under what architectural and governance conditions is Infrastructure Sovereignty feasible in a path-dependent, geopolitically isolated exclave? The framework is developed using Nakhchivan as the primary illustrative context because it combines multiple exclave characteristics in a single case, including geographic separation, external transit dependency, Soviet-era infrastructure, and limited digital development, making it analytically productive for framework building. The resulting framework is proposed as transferable to other geopolitically isolated contexts sharing these structural characteristics, including continental exclaves such as Kaliningrad (Russia) and Ceuta (Spain), and island territories with comparable supply-chain exposure. Transferability is not assumed; it is a hypothesis requiring case-specific Precondition Transferability Assessments, as developed in Section 4.3.

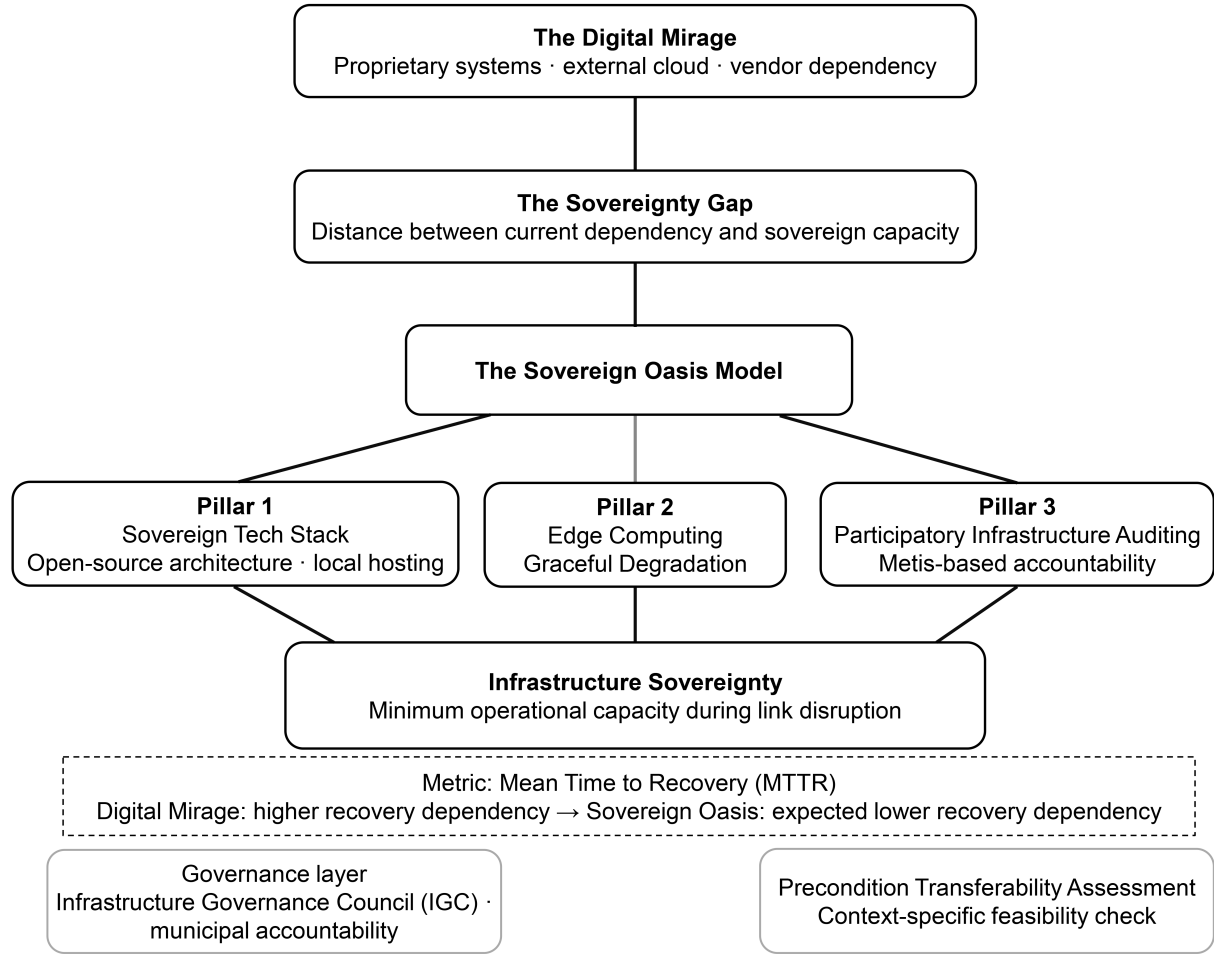


Figure 1. Conceptual progression from the Digital Mirage condition to Infrastructure Sovereignty through the Sovereign Oasis three-pillar architecture, supported by governance mechanisms, Mean Time to Recovery (MTTR) measurement, and Precondition Transferability Assessment

To develop the framework, the paper employs a Most Dissimilar Systems Design (MDSD) comparative analysis across three cases—Aghali Smart Village (Azerbaijan), Tallinn (Estonia), and Nakhchivan City—drawing on high-modernist planning theory [6], polycentric governance [7], and the argument that technical systems embed political arrangements [8]. Mean Time to Recovery (MTTR) is proposed as the primary measurement methodology for future empirical testing. Nakhchivan is used throughout as the illustrative application context; empirical testing of the framework across actual exclave deployments is the explicit agenda for future research.

2 Theoretical Framework

2.1 High Modernism and the Legibility Trap

The intellectual history of smart city failure is, in large part, a history of what Scott [6] called high modernism: the state's compulsion to render complex social reality legible through the imposition of standardised, centrally administered order. Scott's canonical examples, including Soviet collectivisation, Tanzanian villagisation, and Haussmann's Paris, share a common logic: local, practical, tacit knowledge, what Scott terms *metis*, is displaced by the schematic certainties of technocratic planning. The result is a system that may appear coherent on the administrator's dashboard while failing at the granular level of lived experience.

Early accounts of smart cities emphasised the promise of big data, civic hackers, and digitally mediated urban innovation [9]. However, contemporary smart city initiatives can also reproduce this high-modernist dynamic with notable fidelity. Hollands [1] warned early that smart city discourse risks functioning as corporate storytelling rather than urban problem-solving, a critique developed further by Söderström et al. [4] and, more recently, by Sadowski [3], who documents how smart city vendor ecosystems create structural conditions of corporate capture over municipal data infrastructure. Kitchin [2, 10] extends this analysis to show how datafied urban environments embed assumptions about efficiency and optimisation that serve the interests of platform providers rather than city residents. The city optimised for the dashboard is not necessarily the city that serves the person trying to irrigate a field, manage a small business, or access emergency services during a link disruption.

For exclave urban contexts, the risk of reproducing this error carries heightened strategic consequences. Even when well intentioned, a smart city initiative may become a digital high-modernist project in which sensors and dashboards impose legibility while ignoring the *metis* of residents and practitioners who constitute the actual operational fabric of the city. Mattern [11] provides an additional corrective: resilient urban infrastructure is characterised not by the sophistication of its technology but by the robustness of its maintenance culture. A system that no local engineer can repair is not a smart system for an exclave; it is a liability.

2.2 Polycentric Governance and the Digital Commons

The resolution to the high-modernist trap is not the absence of planning but a different architecture of planning. Elinor Ostrom's work on common-pool resource (CPR) governance offers the theoretical basis for this alternative. Ostrom [7] demonstrated that communities are capable of sustainably managing shared resources without either state monopoly or market privatisation, provided that the governance architecture satisfies identifiable design principles, including clearly defined boundaries, rules congruent with local conditions, collective choice arrangements, monitoring mechanisms, graduated sanctions, and recognised rights to self-organise.

Hess and Ostrom [12] extended this framework explicitly to knowledge and digital infrastructure, arguing that open-source software ecosystems and community-maintained digital architectures constitute a form of knowledge commons subject to analogous governance principles. This body of work forms the theoretical basis for what has been termed open-source urbanism: the application of commons-based, citizen-maintainable digital infrastructure to the urban environment, where the goal is not merely technical openness but the civic capacity to sustain, modify, and govern shared urban systems without dependence on proprietary external providers. De Lange and de Waal [13] situate this tradition within a broader argument about citizens' rights to shape their digital urban environment. The term open-source urbanism is used here as an extension of this citizen-engagement literature to the domain of infrastructure governance, rather than as a direct synonym for their original usage. An important distinction applies to the digital commons. Unlike physical CPRs, such as fisheries or irrigation systems, digital resources are largely non-rivalrous, meaning that they can be copied and used simultaneously without depletion. The governance challenge is therefore not preventing over-extraction, but preventing under-contribution and ensuring sustained maintenance. These are challenges that Ostrom's design principles address directly.

It is important to acknowledge that open-source architectures are not without limitations. Open-source systems require active community maintenance. If the local developer pool is too small or insufficiently skilled, the commons may suffer from under-contribution and gradual technical debt accumulation, with potential consequences for software sustainability and quality [14]. Open-source software may also introduce security vulnerabilities when patches are delayed, and software security patch management involves socio-technical challenges that require sustained local capacity [15]. In addition, the absence of a vendor support contract places the full burden of incident response on local institutions. The Sovereign Oasis framework therefore does not treat open-source as an automatic solution. Rather, it treats open-source architecture as a governance design choice that is appropriate only when the necessary institutional conditions can be established and maintained, including a functioning Infrastructure Governance Council (IGC), a trained local engineering community, and sustained political commitment.

A further important consideration concerns the applicability of polycentric governance principles across diverse political and administrative contexts. Ostrom's empirical research documented functioning CPR governance systems across a wide range of political regimes, including irrigation systems in Nepal, Spain, and the Philippines operating

under varying degrees of state centralisation [7]. The design principles she identified do not require democratic governance as a precondition. Instead, they require defined boundaries, congruent rules, and accountability mechanisms, all of which are achievable within centralised administrative structures when institutional will exists. This observation is directly relevant to the exclave context, where governance architectures may differ from liberal-democratic models while still supporting the collective maintenance arrangements required by the Sovereign Oasis framework.

The application to exclave urban digital infrastructure follows directly. A Sovereign Tech Stack maintained by a distributed community of local engineers, university researchers, and municipal technicians constitutes a digital CPR in the Ostromian sense. This framing also resolves a potential internal tension in the argument. If a central state agency designs and controls the sovereign architecture, it may reproduce the high-modernist structure while merely changing the vendor's nationality. Polycentric governance, when distributed, locally maintained, and collectively accountable, can help avoid this problem. The people maintaining the code may, in many cases, be the same people whose water and power systems the code manages, which is the mechanism through which *metis* enters and sustains the system.

2.3 Artefacts, Politics, and the Sovereignty Gap

The concept of Infrastructure Sovereignty, as used in this paper, builds on Winner's foundational argument that artefacts have politics, meaning that technological systems embed and reproduce particular social and political arrangements [8]. Winner's insight provides the normative basis for treating infrastructure dependency as a strategic rather than merely technical problem. For example, a city that can manage its own traffic lights only by calling a vendor's support line has ceded a dimension of political autonomy that cannot be recovered through regulatory means alone. This paper does not present Infrastructure Sovereignty as a wholly new theoretical framework. Rather, it applies Winner's foundational insight to the specific and undertheorised context of exclave urban infrastructure, where external dependency carries heightened strategic consequences.

The Sovereignty Gap is defined operationally as the measurable distance between a city's current dependency profile and the condition of sovereign operational capacity. This refers to the ability of a local administrative body to maintain, modify, and secure core urban functions without recourse to external proprietary software or foreign-hosted cloud dependencies. Recent scholarship reinforces the importance of this distinction. Müller et al. [16] demonstrate that smart city trajectories are deeply path-dependent and shaped by the legacy infrastructure and institutional arrangements of specific urban contexts. Mullick and Patnaik [17] further document how urban communities can be systematically excluded from the benefits of smart city investment when governance frameworks prioritise vendor efficiency over local capacity. The process-tracing illustration in Section 6 makes this gap visible at identifiable nodes in a representative smart grid dependency chain.

2.4 The Urban Planning Dimension

The Sovereign Oasis framework connects directly to established urban planning concerns that extend beyond digital governance. At the land-use planning level, the siting of edge computing nodes, on-premise server infrastructure, and distributed sensor networks requires integration into local development plans and zoning frameworks. These are decisions that fall squarely within the remit of municipal planning departments. The framework's resilience architecture is not simply a technology layer. It is a spatial infrastructure that demands planning-led coordination across districts, neighbourhoods, and rural-urban interfaces.

At the level of infrastructure planning, the retrofit challenge discussed in Section 4.4 connects directly to the problem of integrating digital systems into existing water, power, and transport networks, which is a core concern of urban infrastructure planning theory [18]. The Sovereign Tech Stack's localised data hosting layer has direct implications for urban capital investment planning. Decisions about where to locate on-premise servers, how to route fibre connectivity within the exclave, and how to phase the Internet of Things (IoT) sensor deployment across legacy analogue networks all require coordinated capital programming over multi-year budget cycles. From a resilience planning perspective, the Graceful Degradation protocol is directly analogous to the priority-based service continuity frameworks used in conventional emergency urban planning. The framework formalises and extends this logic for digitally managed infrastructure. Finally, the Participatory Infrastructure Auditing mechanism described in Section 5.3 represents a digital extension of community engagement practices already embedded in participatory urban planning traditions [19, 20]. It is adapted for the technical reporting requirements of infrastructure governance rather than spatial policy deliberation.

3 Methodology

3.1 Most Dissimilar Systems Design

This paper employs a MDSD comparative methodology to develop and stress-test the theoretical framework. MDSD is appropriate when the researcher seeks to demonstrate that a relationship between variables is coherent

across cases that differ substantially in background conditions [21]. The three cases selected differ in governance context, infrastructure legacy, geographic situation, and economic base, yet all illuminate different dimensions of digital urban governance under conditions of external vulnerability.

Case one—Aghali Smart Village (Azerbaijan)—is a new-build smart settlement constructed in the post-conflict Zangilan region with high state investment and with substantially fewer legacy-infrastructure constraints than retrofit urban centres. It serves as a greenfield feasibility reference. Case two—Tallinn, Estonia—represents a small, geopolitically exposed state that has constructed a sophisticated distributed digital governance architecture in explicit response to external disruption threats. It serves as the institutional sovereignty benchmark. Case three—Nakhchivan City—is a legacy urban centre with Soviet-era infrastructure and acute geopolitical isolation. It serves as the primary illustrative application context for the framework.

3.2 Analytical Approach: Process-Tracing Illustration

The primary analytical method is a process-tracing illustration of the dependency chain of a single representative smart service: smart grid energy management. It is important to clarify the epistemic status of this illustration. Causal Process Tracing in its full methodological sense, as developed by Beach and Pedersen [22], requires empirical tracing of a real causal mechanism through observable intermediate steps in an actual case. As a theoretical framework paper, this study instead offers a structured architectural comparison of two hypothetical deployment models, the Digital Mirage and the Sovereign Oasis, to make the Sovereignty Gap visible at each node in the dependency chain. This is a process-tracing illustration used to develop and clarify the framework's analytical logic, rather than a full causal process-tracing analysis of an empirical case. Empirical process tracing is proposed as the priority method for future research.

The causal mechanism proposed by the framework is that proprietary vendor lock-in at the software layer creates Sovereignty Gaps at identifiable nodes in the infrastructure chain, making urban services vulnerable to external disruption independent of the physical integrity of local hardware. The primary alternative explanation, namely that Sovereignty Gaps result from regulatory failure rather than technical architecture, is partially addressed through the Tallinn comparison. Full adjudication between these mechanisms requires empirical primary data, which is identified as the central agenda for follow-up research in Section 8.

3.3 Mean Time to Recovery as a Measurement Methodology

MTTR is proposed as the primary resilience measurement methodology for future empirical testing of the Sovereign Oasis framework. Commonly used in reliability engineering and site reliability engineering, MTTR measures the time required to restore a system or service after a failure and is adapted here for application to urban infrastructure systems [23]. MTTR measures the average time required to restore a critical urban service following a disruption event. The framework's core prediction is that Sovereign Oasis architectures will produce systematically lower MTTRs than Digital Mirage architectures, because restoration under sovereign conditions depends on local technical capacity rather than external vendor availability. Lu et al. [24] provide empirical support for the broader association between digital infrastructure construction and urban resilience outcomes.

For the purposes of this theoretical framework, MTTR is used as a directional measurement logic rather than as a source of specific numerical benchmarks. The architectural comparison in Section 6 applies this logic qualitatively, while the empirical measurement of baseline and post-implementation MTTR in actual exclave deployments is proposed as the primary quantitative task for future research, as further discussed in Section 8.

4 Comparative Analysis

4.1 Contextual Profile: Nakhchivan Autonomous Republic

Table 1 provides a contextual profile of the Nakhchivan Autonomous Republic, highlighting the geographic, demographic, infrastructural, and administrative conditions relevant to the illustrative application of the proposed framework. The demographic, territorial, and related contextual information is compiled from publicly available statistical sections of the Azerbaijan Statistical Information Service (ASIS) of the State Statistical Committee of the Republic of Azerbaijan [25].

4.2 Case One: Aghali as a Greenfield Feasibility Reference

The Aghali Smart Village project in the Zangilan district was developed as part of Azerbaijan's post-conflict reconstruction programme and has been presented in official sources as the country's first smart village project [26]. It provides a greenfield reference case in which digital infrastructure can be planned from the initial design stage, rather than retrofitted into pre-existing urban systems. Compared with legacy urban centres, such a new-build settlement is likely to face fewer constraints related to ageing infrastructure, fragmented utility networks, and incompatible analogue systems.

The project has been reported to incorporate ecological housing, smart lighting, waste management infrastructure, alternative-energy-supported agricultural production, digitalised water supply, and technology-supported agricultural monitoring [27]. Food and Agriculture Organization of the United Nations (FAO)’s Digital Villages Initiative profile also identifies Aghali as a digital village context in which agricultural livelihoods, digital skills, and access to technology remain central development issues [28]. These features make Aghali relevant to the present framework because they indicate that several technical components associated with digitally managed infrastructure can be deployed within the Azerbaijani institutional and geographic context. However, publicly available information about Aghali remains largely based on official communications, project descriptions, and policy-oriented sources, while independent performance evaluation remains limited. Therefore, this case is used here as a feasibility reference rather than as a fully verified empirical model.

The relevance of Aghali to the Sovereign Oasis framework lies mainly in its contrast with retrofit exclave contexts. In a greenfield setting, digital systems and physical infrastructure can be designed together from the outset, which may reduce integration barriers. In a retrofit setting such as Nakhchivan City, by contrast, new digital systems must be connected to existing water, power, transport, and administrative infrastructures. These integration points may become Sovereignty Gaps when the software interfaces, diagnostic tools, or maintenance protocols are controlled by external vendors. The Aghali case therefore helps clarify the path-dependency problem that the Sovereign Oasis framework seeks to address.

Table 1. Contextual characteristics of the Nakhchivan Autonomous Republic

Dimension	Key Data and Characteristics
Territory and area	Nakhchivan Autonomous Republic (NAR); approximately 5,500 km ² ; an exclave of Azerbaijan
Population	Approximately 458,900 for the NAR and approximately 94,400 for Nakhchivan City, based on the 2019 population census
Geographic isolation	Separated from mainland Azerbaijan by Armenian territory; bordered by Iran to the south, Turkey to the west, and Armenia to the north and east
External connectivity	Dependent on external transport corridors and cross-border transit routes for land connectivity; served by Nakhchivan International Airport
Infrastructure legacy	Legacy water, power, and road infrastructure partly inherited from the Soviet period, with partial modernisation since 1991
Economic base	Agriculture, including viticulture and cotton, together with salt extraction and a limited industrial base
Digital development status	Public administration digitalisation is developing; however, available public sources provide limited evidence of a dedicated large-scale smart city programme in Nakhchivan
Administrative structure	Autonomous republic within Azerbaijan, with regional representative and executive institutions operating under the constitutional framework of Azerbaijan
Higher education	Nakhchivan State University, established in 1967, serves as a major regional higher education institution with engineering and technical programmes relevant to local capacity development

Sources: Azerbaijan Statistical Information Service (ASIS), State Statistical Committee of the Republic of Azerbaijan [25]. The table is intended to provide an illustrative contextual profile; primary fieldwork verification is required before policy application.

4.3 Case Two: Tallinn as Institutional Sovereignty Design

Estonia’s digital governance architecture, centred on the X-Road distributed data exchange layer, illustrates a well-developed model of Infrastructure Sovereignty at the national scale [29]. X-Road is an open-source, centrally managed, distributed data exchange layer that enables public and private information systems to exchange data securely and in a standardised manner [30]. Its architecture does not require a single central database. Instead, data remain with participating organisations and are exchanged through a secure interoperability layer. This architecture has been progressively strengthened since 2001 and complemented by the Data Embassy concept, under which critical state data and services are additionally hosted on server resources outside Estonia’s territorial boundaries as a continuity measure [31].

A clarification is necessary regarding the Data Embassy model and its relationship to the Sovereign Oasis framework proposed here. Estonia’s strategy involves hosting state-controlled server resources abroad as a redundancy measure against domestic territorial disruption. This differs from the local-hosting sovereignty model proposed for exclaves. The two approaches are not contradictory; rather, they reflect different threat models. Estonia’s primary

concern is that domestic disruption may render domestic servers inaccessible. An exclave's primary concern is that external link severance may render foreign-hosted servers inaccessible. The appropriate sovereignty architecture therefore differs by threat model. Exclave contexts require on-premise local hosting because external connectivity, rather than domestic territorial control, is the primary vulnerability.

Applying Ostrom's design principles to the Estonian case suggests why X-Road has achieved institutional durability. Relevant factors include clearly defined access boundaries, rules aligned with Estonia's legal framework, institutional coordination for system governance, and recognised rights and responsibilities for maintaining individual nodes. The transferable lessons are institutional rather than purely technical. Sovereignty is a legislative and organisational commitment before it is a hardware purchase. Distributed architecture may outperform centralised architecture under threat conditions relevant to small or geopolitically exposed entities. However, the Estonian case must be read in its specific context, including more than two decades of European Union (EU) membership, sustained national investment over more than twenty years, and highly developed digital-government capacity [32]. These preconditions, detailed in Table 2, do not transfer automatically. The lesson from Tallinn is therefore less about copying the technology than about understanding the institutional commitment that makes the technology sustainable.

Before applying Estonia's lessons to any exclave context, a structured Precondition Transferability Assessment is required. Table 2 maps Estonia's enabling preconditions against the illustrative conditions of Nakhchivan as an application of this assessment methodology.

Table 2. Precondition Transferability Assessment from Tallinn to Nakhchivan

Precondition	Estonia/Tallinn	Nakhchivan	Transferability Assessment
Duration of investment	25+ years of sustained investment since 1997	No publicly documented long-term digital-governance programme of comparable scale	Low; long-term institutional commitment is required
Legal framework	Digital Signatures Act and Public Information Act are in place	A comparable digital-governance legal framework is not clearly documented	Medium; legal development is primarily a policy and institutional task
Digital literacy among civil servants	Highly developed digital-government capacity	University-educated cohort potentially available; the regional university may serve as an anchor institution, pending capacity assessment	Medium; structured capacity-building would be required
Primary threat model	External military aggression and cyberattacks on domestic infrastructure	Supply-chain disruption and external corridor severance	Different; local hosting is more relevant than foreign-hosted redundancy
Governance structure	Multi-stakeholder foundation model supported by formal institutional arrangements	Centralised administrative tradition	Low to medium; a phased and context-specific governance model would be required
Technology talent pool	Relatively developed and internationally connected digital sector	Limited; local talent retention is a structural challenge	Low; talent development would need to be built into the framework
Open-source and interoperability capacity	Open-source and interoperability principles are embedded in digital-government practice	Dedicated open-source and interoperability arrangements are not clearly documented	Medium; achievable through legislative and institutional design

Note: Azerbaijan Statistical Information Service (ASIS), State Statistical Committee of the Republic of Azerbaijan [25]. Empirical verification is required before policy application.

The transferability assessment suggests that Estonia's institutional architecture cannot be transplanted directly into exclave contexts. What transfers are its design principles, including distributed architecture, open-source foundations, and multi-stakeholder maintenance governance. These principles must be adapted to governance environments with different preconditions and threat models. This distinction between transplanting architecture and transferring principles is central to the Sovereign Oasis framework.

4.4 Case Three: Nakhchivan as a Retrofit Challenge

Nakhchivan City serves as the framework's primary illustrative application context. The analysis does not claim to provide an empirically verified account of Nakhchivan's infrastructure conditions. Rather, it illustrates how the framework's analytical categories, including the path-dependency gap, the Sovereignty Gap, and the Precondition Transferability Assessment, may apply to a representative exclave case. The contextual data in Table 1 establish the basic parameters of this setting, while empirical verification through primary fieldwork is identified as a priority for future research in Section 8.

Path-dependency constraints in legacy exclave urban centres are likely to be significant, particularly where infrastructure systems were developed under earlier centralised planning and analogue management conditions. Water networks, electrical distribution systems, transport infrastructure, and building stock were generally not designed for sensor integration or real-time digital management. Retrofitting IoT systems into such networks requires interfaces between legacy analogue systems and new digital layers, and these interfaces may become Sovereignty Gaps when they depend on proprietary software, vendor-controlled diagnostic tools, or externally managed maintenance protocols. Where interface software is proprietary, local engineers may be unable to diagnose or repair failures without vendor access. Marvin et al. [33] document this vendor-capture dynamic across smart city deployments, while Meijer and Thaens [34] show that technological and social legacies create path dependencies that shape smart city development, helping explain why legacy infrastructure conditions can constrain the scope and pace of smart city retrofitting in established urban centres.

The geopolitical constraints of exclave infrastructure are illustrated by Nakhchivan's dependence on external transport corridors and cross-border transit routes for land connectivity and supply-chain access. In the present framework, such external linkages are treated as uncertain planning conditions. They may be available under normal circumstances but cannot be assumed to remain continuously reliable over long infrastructure planning horizons. Therefore, smart systems whose operation depends on uninterrupted external connectivity may be poorly suited to exclave contexts. This does not mean that external connectivity should be avoided. Rather, it means that critical urban functions should be capable of maintaining minimum service levels when external links are disrupted.

The Nakhchivan case also illustrates the potential institutional assets available in some exclave contexts. A regional university may serve as a potential anchor for distributed technical maintenance, although this would require verification of its engineering capacity, institutional willingness, and long-term resource base. Municipal utility practitioners may possess tacit knowledge of existing infrastructure that external vendors cannot readily replicate, but the depth and accessibility of this knowledge also require empirical assessment. Local agricultural and industrial communities may have economic interests in digitally managed water, energy, and resource systems, which could create constituencies for maintenance and sustainability. However, these assets do not automatically constitute a polycentric governance structure. A formal institutional architecture is still required to connect, coordinate, and sustain them. The IGC design proposed in Section 5 offers one possible template for such an architecture.

5 The Sovereign Oasis: A Three-Pillar Resilience Architecture

5.1 Pillar One: The Sovereign Tech Stack

The Sovereign Tech Stack is a layered technical architecture in which each layer is maintained by identifiable local actors using open-source tools. Its purpose is to reduce proprietary lock-in across the dependency chain. The architecture consists of three layers: an open-source kernel layer, including operating systems, data management protocols, and sensor firmware based on open standards; a localised data hosting layer, consisting of on-premise servers physically located within the exclave's administrative territory; and a community-maintained API layer, based on open protocols for inter-system communication that can be documented and modified by local developers without vendor permission.

Governance of the Sovereign Tech Stack is formalised through an IGC. The IGC is proposed as a multi-stakeholder body comprising representatives from the regional university's engineering faculty, engineers holding professional certifications under applicable national frameworks, and vetted local technology practitioners. Their qualifications and terms of participation would be defined in the IGC's founding ordinance. Its legal basis would be a municipal ordinance or administrative regulation establishing the IGC's mandate, membership criteria, vetting procedures, and decision-making protocols. Decisions regarding system modifications, access permissions, and maintenance priorities would be made through a documented collective-choice process requiring a defined quorum. Outcomes would be recorded as formal administrative decisions and made available through the relevant municipal authority where appropriate. The IGC's primary accountability mechanism would be a published annual infrastructure performance report reviewed by the relevant municipal authority. This structure applies Ostrom's design principles, including defined boundaries, collective-choice arrangements, monitoring, and accountability, in a form compatible with centralised administrative governance structures. This is consistent with the observation in Section 2.2 that polycentric maintenance governance does not necessarily require democratic preconditions.

The Sovereign Tech Stack represents a political choice as much as a technical one. It treats urban digital infrastructure as a shared governance resource rather than merely as a purchased service. This choice affects who can repair, modify, and maintain the system when it fails, which is a strategic issue in exclave contexts.

5.2 Pillar Two: Edge Computing and Graceful Degradation

Edge computing refers to the processing of data at or near the source, such as the sensor, meter, agricultural node, or local hub, rather than routing all data to a distant cloud server [35]. In a conventional smart city architecture, a query about soil moisture may require data to leave the exclave, reach an externally hosted server, be processed remotely, and then return a response. A failure at any point in this external chain can interrupt the service loop.

In the proposed edge computing architecture, the same query is processed locally within the sensor cluster, local hub, or exclave-level edge server. External internet connectivity may still be used for updates, calibration, and non-critical data aggregation. However, the critical service loop, including the functions required to keep water, power, and agricultural systems operating, remains within the exclave's network perimeter. In this architecture, the system is designed to benefit from connectivity when it is available while maintaining minimum operational capacity when external links are disrupted.

Graceful Degradation is the operational principle that governs system behaviour during link severance. Instead of failing catastrophically when external connectivity is lost, the system maintains minimum viable service across critical functions by reducing or suspending non-critical services in a predefined sequence. Three governance decisions should be resolved and documented before deployment rather than improvised during a crisis. First, the priority schedule should be authorised by the IGC through a collective-choice process and recorded as a formal administrative decision. Second, the technical threshold that triggers degradation mode should be defined, codified, and testable. Third, the monitoring and verification protocol should be assigned to a named operational role within the IGC structure, with a defined escalation pathway.

5.3 Pillar Three: Participatory Infrastructure Auditing

The third pillar is a Participatory Infrastructure Auditing (PIA) mechanism proposed in this paper. It draws on participatory planning and complex governance literature [19, 20], as well as participatory urban sensing and mobile reporting research [36] and adapts these ideas to the specific task of infrastructure maintenance in geopolitically isolated exclaves. It does not require deliberative democratic governance structures. Instead, PIA allows residents to report infrastructure failures directly to a structured log maintained by the IGC. Figure 2 illustrates the five-stage information flow through the PIA system.

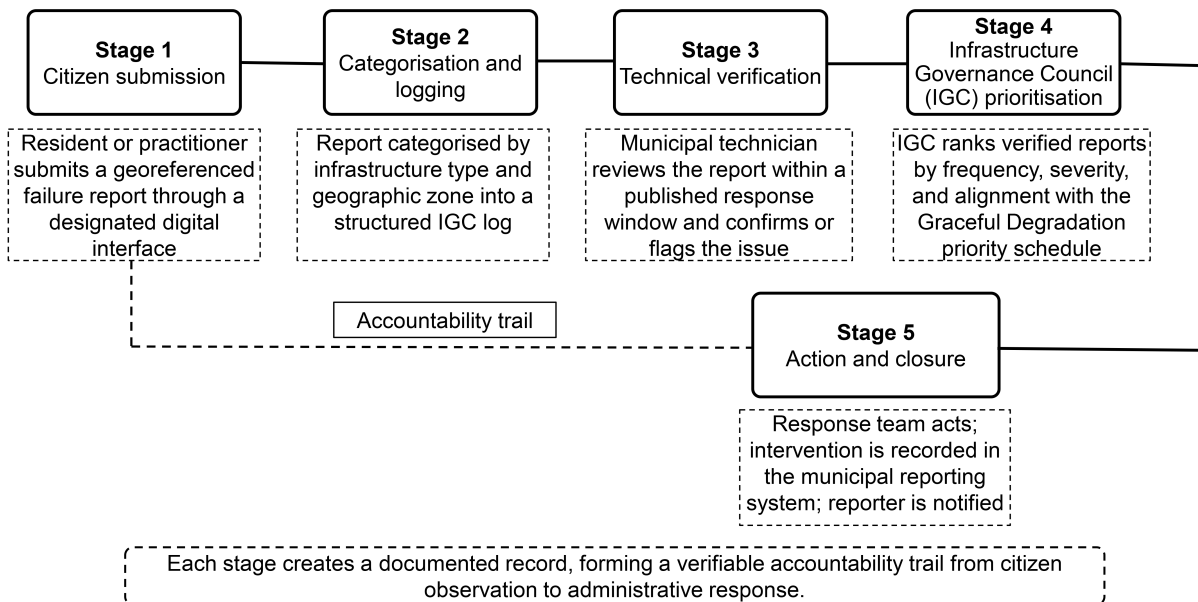


Figure 2. Information flow in the Participatory Infrastructure Auditing system, showing the progression from citizen submission to administrative response and closure

The mechanism is designed to capture metis, namely the tacit local knowledge of infrastructure behaviour that practitioners and residents may possess before anomalies appear in administrative monitoring systems. In Stage 1, a resident or practitioner submits a georeferenced failure report through a designated digital interface. Where available,

existing municipal administrative credentials may be used for identity verification. In Stage 2, the report is categorised by infrastructure type and geographic zone and entered into the structured IGC log. In Stage 3, a designated municipal technician reviews and verifies the report within a published response window, confirming the issue or flagging it as a duplicate or misclassification. In Stage 4, the IGC prioritises verified reports according to frequency, severity, and alignment with the Graceful Degradation priority schedule. In Stage 5, the assigned response team acts on the report, records the intervention in the municipal reporting system, and notifies the original reporter, thereby closing the accountability loop.

This five-stage cycle creates a documented accountability trail from citizen observation to administrative response. Identity verification should rely on existing administrative systems where possible, avoiding the creation of new surveillance infrastructure. The mechanism is designed to function within centralised administrative environments by routing participation through technical reporting rather than political deliberation.

5.4 Framework Integration with Existing Urban Governance Structures

A critical question for the practical applicability of the Sovereign Oasis framework is how its proposed mechanisms can be integrated into existing municipal governance structures rather than operating in parallel to them. The framework is not designed to replace existing administrative arrangements. It is intended to be embedded within them as a formalised technical governance layer. The IGC design presented here represents one possible governance architecture rather than the only viable model. Alternative arrangements, such as embedding digital infrastructure governance within an existing municipal utility commission, delegating it to a university-linked technology transfer office, or establishing a public-private partnership with explicit sovereignty safeguards, may be more appropriate in specific institutional contexts. The IGC model is therefore offered as a theoretically grounded starting point. Its specific form should be adapted through stakeholder consultation and piloting in each exclave context.

In practice, the IGC would function as a standing technical committee within the existing municipal governance structure, reporting to the relevant executive body responsible for public utilities and infrastructure. Its founding ordinance would be issued by the municipal executive as a normal administrative act. The IGC's annual performance report would feed into the municipal budget cycle, with infrastructure priorities identified through the PIA mechanism informing capital expenditure decisions for the following fiscal year. This connection between citizen-reported failure data, IGC prioritisation, and municipal budget allocation is where the framework's governance logic connects most directly to routine urban management practice.

For public service provision, the Sovereign Tech Stack's open-source architecture would enable municipal service departments, including water, power, waste management, and emergency services, to access shared infrastructure data through the community-maintained API layer without proprietary licensing requirements. Each department would maintain write access to its own service data and read access to relevant cross-sector data, with access permissions governed by the IGC's collective-choice protocols. For city-level digital management, the edge computing model would ensure that the exclave's operational data remain within its administrative territory, enabling local analytics and reporting without routing sensitive urban systems data through external servers. Integrating this architecture into city-level digital management plans would require coordination with the land-use and infrastructure planning functions described in Section 2.4.

6 The Smart Grid Dependency Chain: A Theoretical Illustration

This section develops a structured architectural comparison of two smart grid deployment models in order to illustrate the Sovereignty Gap in the exclave context. Consistent with the methodological clarification in Section 3.2, the comparison is presented as a theoretical illustration rather than as an empirical tracing of an observed failure event.

6.1 Model A: The Digital Mirage Architecture

In a conventional proprietary smart grid deployment, a residential solar panel may be metered by a device running proprietary firmware. The meter may report to a local aggregator running vendor-controlled software, which then transmits data to a cloud-hosted regional data centre located outside the exclave's administrative territory. The grid management system processes the data and issues routing instructions back to local switching hardware. Any modification to the system logic may require an active vendor support contract and externally controlled access credentials.

Dependency chain analysis identifies five distinct Sovereignty Gaps in this architecture: the meter firmware (proprietary, not modifiable without a vendor licence); the aggregator software (vendor-maintained, requiring an active support contract for updates); the cloud data centre (externally hosted, subject to the legal jurisdiction of the host country and the availability of the external link); the management system logic (inaccessible to local engineers without vendor access credentials); and the switching hardware protocols (proprietary, requiring vendor-supplied diagnostic tools for maintenance). A severe external link disruption could affect several of these nodes at the same time, reducing system functionality even when local physical hardware remains intact.

6.2 Model B: The Sovereign Oasis Architecture

In the proposed sovereign architecture, the same residential solar panel is metered by a device running open-source firmware maintained by locally authorised technical personnel. Data are processed at an edge computing node located within the exclave, such as a local server maintained by municipal technicians, university engineers, or other approved technical actors. The grid management logic runs on documented and modifiable open-source software. Switching hardware uses open protocols compatible with locally available diagnostic tools. External internet connectivity carries only non-critical telemetry, rather than data required for basic grid function.

Under this architecture, a total external link severance would trigger Graceful Degradation. The grid management system would switch to local optimisation mode and prioritise critical loads according to the pre-published IGC priority schedule. Restoration would depend primarily on local technical response capacity rather than vendor support queues or the transit time for imported components. This distinction is the structural difference that the framework's MTTR methodology is designed to assess empirically.

Table 3. Theoretical dependency profile of Digital Mirage and Sovereign Oasis architectures

Dependency Node	Model A: Digital Mirage	Model B: Sovereign Oasis
Meter firmware	Proprietary; vendor licence required for modification	Open-source; modifiable by locally authorised technical personnel
Data processing	Cloud-hosted and subject to external legal and connectivity conditions	Edge computing; on-premise processing within the exclave territory
Management logic	Vendor-controlled; active support contract may be required	Open-source; documented and locally maintainable
Hardware protocols	Proprietary; vendor diagnostic tools may be required	Open standards; compatible with locally available diagnostic tools
Link severance impact	Potential cascade failure across multiple dependency nodes	Graceful Degradation to local-only operational mode
MTTR for critical node	Expected to be higher because of vendor dependence	Expected to be lower if sufficient local technical capacity exists
Governance accountability	External vendor service-level agreement with limited local enforcement	Internal Infrastructure Governance Council (IGC) accountability with locally defined service procedures

Note: Mean Time to Recovery (MTTR) predictions are directional and qualitative. Empirical measurement of actual MTTR values in exclave deployments is proposed as a primary task for future research.

7 Scope and Limitations

As a theoretical framework paper, this study prioritises conceptual development and comparative architectural reasoning over single-case empirical verification. The framework's application to Nakhchivan is illustrative. It demonstrates how the framework's categories may apply to a representative exclave context, rather than confirming those applications through primary empirical data. This is an acknowledged design choice that reflects the paper's framework-building purpose rather than a claim of empirical validation.

Four limitations bound the claims of the analysis. First, geopolitical fluidity. Exclave external linkages are treated as uncertain planning conditions. They may be present under baseline conditions, but they cannot be assumed to remain permanently stable. The framework provides architectural principles for designing under corridor uncertainty, but it does not predict the political dynamics that determine external access in any specific case.

Second, MTTR verifiability. The MTTR predictions in Table 3 are directional and qualitative rather than quantitative estimates. The framework predicts that sovereign architectures may produce lower MTTRs than dependent architectures under appropriate institutional and technical conditions. However, it does not specify the magnitude of that difference without empirical measurement. Measuring baseline and post-implementation MTTR in actual exclave deployments is therefore a primary quantitative task for future research.

Third, transferability specificity. The Precondition Transferability Assessment in Table 2 maps Estonia's enabling conditions against Nakhchivan as an illustrative exercise. Before applying the framework to other exclave contexts, such as Kaliningrad, Ceuta, or island territories with comparable supply-chain exposure, a fresh assessment is required. The Sovereign Oasis model is a design template for a specific and undertheorised category of urban context, not a universal smart city framework.

Fourth, human capital dependency. The Sovereign Oasis framework depends on the availability of a local engineering community capable of maintaining open-source infrastructure, including university-trained engineers, municipal technicians, and local technology practitioners. In contexts where this capacity does not yet exist, or where

talent retention is severely constrained by economic migration, the framework's operational predictions may not hold regardless of the technical architecture adopted. This is a structural constraint that the framework acknowledges but does not resolve. Building the human capital base required for sovereign architecture is a prerequisite that may take years to develop and depends on broader economic and educational policies beyond the scope of digital urban governance alone. Future research should assess the minimum viable human-capital threshold below which the Sovereign Oasis model cannot function as designed, and should explore capacity-development pathways appropriate to exclave institutional contexts.

8 Conclusion and Future Research

This paper has argued that the dominant model of smart city development, described here as the Digital Mirage, is structurally ill-suited to geopolitically isolated exclaves. By concealing vendor dependency behind a facade of technological sophistication, the Digital Mirage may transform a city's digital infrastructure from an asset into a strategic liability when external conditions deteriorate. The Sovereign Oasis framework developed in this paper proposes a theoretically coherent alternative organised around three pillars: a Sovereign Tech Stack governed through polycentric principles, an Edge Computing model designed to support Graceful Degradation during link severance, and a Participatory Infrastructure Auditing mechanism grounded in local knowledge.

The comparative analysis suggests the theoretical coherence of the framework across three dissimilar cases. Aghali illustrates the feasibility of deploying selected smart-infrastructure components within the relevant geographic and institutional context. Tallinn illustrates the importance of sustained institutional commitment, legal support, and digital-governance capacity in building Infrastructure Sovereignty. The Precondition Transferability Assessment further shows that such lessons cannot be transferred automatically. They require systematic adaptation to the specific threat model, governance structure, and capacity conditions of each exclave context.

The paper's central conceptual contributions include the Digital Mirage, the Sovereign Oasis, the Sovereignty Gap, and the principle that critical urban systems in exclaves should maintain minimum operational capacity when external connectivity is disrupted. These concepts are offered as theoretical building blocks for further research on exclave urbanism, smart infrastructure governance, and urban resilience under conditions of geopolitical isolation.

Three priorities are proposed for future research. First, primary empirical testing should be conducted in Nakhchivan through semi-structured interviews with municipal engineers, infrastructure practitioners, university-based technical experts, and relevant administrative actors. This should be combined with the analysis of available policy documents, infrastructure audit records, municipal utility records, and baseline MTTR measurements in existing water distribution and power management systems. The framework would require substantial revision if empirical testing showed that local maintenance capacity is insufficient to improve recovery performance regardless of architectural design. Such a finding would suggest that the binding constraint is human capital rather than technical architecture, and would redirect the research agenda toward capacity development.

Second, comparative evaluation should be extended to other exclave or similarly isolated contexts, such as Kaliningrad, Ceuta, and island territories with comparable supply-chain exposure. Each application should begin with a fresh Precondition Transferability Assessment rather than assuming that the framework can be directly transplanted across cases. MTTR can serve as a shared comparative metric for assessing whether more sovereign infrastructure architectures improve recovery capacity across different governance environments.

Third, longitudinal assessment is needed to test the framework's directional predictions over time. In any exclave that implements elements of the Sovereign Oasis architecture, data should be collected at baseline, one year after implementation, and three years after implementation. This would allow researchers to assess both immediate and sustained effects on infrastructure resilience, local maintenance capacity, and service recovery performance.

Nakhchivan is used throughout this paper as the illustrative context because it brings together several conditions that are analytically important for framework development, including geographic isolation, external transit dependency, legacy urban infrastructure, and potential institutional assets at the regional university level. Whether it ultimately proves to be an empirically successful application of the framework remains a question for future research based on primary data. The framework is designed to be testable, and its empirical validation should be the next step.

Data Availability

No primary data were collected. The analysis is based on publicly available secondary sources and theoretical case illustration.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The author declares that Claude by Anthropic was used only for language polishing, improving the organization of the text, and enhancing readability. The tool was not used to generate data, results, references, or scholarly conclusions. All content was carefully reviewed and verified by the author, who takes full responsibility for the accuracy, originality, and integrity of the manuscript.

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